

Title: Capacitive biosensor for detection of GHB

5 The invention relates to a capacitive biosensor for detection of GHB, a device comprising said biosensor and a method for detecting GHB.

Background

10 In recent years, the usage of *gamma*-hydroxybutyrate, commonly known as GHB, for drugging in nightlife contexts has increased noticeably. GHB spiking is an instance where droplets of the said drug are added to a person's drink without them knowing. This can be to increase their vulnerability to rob them, to commit sexual assault, or for other reasons. GHB is the most commonly used 'rape drug' in Western Europe. It has a fast onset of action, and in excessive doses or when mixed with
15 alcohol it can cause permanent damage. Symptoms of a high dosage of GHB can be apnea, heart attack, cardiac arrest, coma, and blood pressure problems, which can result in death. (<https://www.drugs.com/illicit/ghb.html>; accessed on 6 September 2022).

A large portion of the Western European population says they have had
20 experience with their drink being spiked, or otherwise knew someone whose drink was poisoned. A European survey of sexism-free-night reported that out of the around 4500 people who filled in the survey, 7.8% thought they had been drugged through "Premeditated [means] or with prior preparation", meaning someone made them consume a drug without them being aware of it (Plaza, L., Ferrer, R. & Vale Pires, C.
25 (2022). Sexism Free Night – Research Report. Sexism Free Night European Project. Available online at: www.sexismfreenight.eu).

Another research showed that in the UK, one in nine women thinks their drink has been spiked at least once, and 28% of the interviewed people know someone whose drink is suspected to have been poisoned (<https://yougov.co.uk/topics/lifestyle/articles-reports/2021/11/18/one-nine-women-have-been-drink-spiked>;
30 accessed on 6 September 2022).

In the Netherlands, NOS handed out a survey to 5000 people; 5% of people who filled out the survey said they had been spiked, and more than 50% of questioned women take the possibility of being drugged into account when going out

(<https://nos.nl/artikel/2428397-overal-berichten-over-drogeringen-in-uitgaansleven-hoe-zit-dat>; accessed on 6 September 2022).

Portable tests for detection of GHB are known, such as the DetectaChem GHB test, but are expensive, single-use, and slow

(<https://www.detectachem.com/product/ghb-drug-test/> accessed on 27 September 2022). For this reason, such portable tests are rarely used on-site, such as in a bar, nightclub or the like.

Further, hospitals in the Netherlands use Gas Chromatography-Flame Ionization Detection (GC-FID) or High-Performance Liquid Chromatography (HPLC) procedures that require large devices operated by professionals. This type of detection delays the detection process and lowers the efficiency of the emergency response.

Accordingly, there is a need for fast, on-site, accurate GHB testing device to prevent the consumption of the spiked drink, learn more about the frequency of GHB spiking, and enhance the emergency response.

The inventors advantageously realized that with a capacitive biosensor one or more of the above-mentioned drawbacks can be overcome.

Accordingly, the invention relates to a capacitive biosensor for detecting *gamma*-hydroxybutyric acid (GHB), a derivative thereof or a salt thereof, comprising a first substrate, a nucleic acid molecule immobilized on said first substrate and a transcription factor specifically bound to said nucleic acid molecule, wherein said transcription factor comprises a first binding site comprising the amino acid sequence of SEQ ID NO: 1 or a functional derivative thereof.

Figures

Figure 1. Visualization of the average capacitance during the vector impedance measurements. The capacitance of the electrode A (with BlcR binding site on DNA) and electrode B (without BlcR binding site on DNA) has been measured without BlcR or SSA addition, labeled 'DNA', after incubation with BlcR, labeled 'BlcR', and after incubation with SSA, labeled 'SSA'. The standard deviation has also been included.

Figure 2: A picture of the UTI board.

Figure 3: A diagram of the electrical system of the UTI. On the left side there are three capacitors that are outside of the main board. The system first uses capacitance-to-voltage converter to convert the capacitance to a voltage (CVC). A voltage-to-period converter is then used to translate this voltage signal into a

particular period (VPC). Lastly, the frequency is adjusted to the required data acquisition frequency using a digital divider.

Figure 4: A diagram showing the connections of the UTI. Five capacitors are represented in the figure: C_{BA} is 0, C_{CA} is our reference capacitor, and C_{DA} is the capacitor we are using in the measurements. For our device we don't use C_{EA} and C_{FA} .

Figure 5: The overview of the settings used for the UTI measurements.

Figure 6. Visualization of the average capacitance during the UTI measurements. The capacitance of the electrode A (with BlcR binding site on DNA) and electrode B (without BlcR binding site on DNA) has been measured without BlcR or SSA addition, labeled 'DNA', after incubation with BlcR, labeled 'BlcR', and after incubation with SSA, labeled 'SSA'.

Figure 7: Simplified overview of the biological system immobilized on the interdigitated electrodes in the scenarios in which GHB is not present and when it is present.

Figure 8: Visualization of the placement of the water molecules in the scenarios in which GHB is not present and when it is present.

Detailed description

The term 'or' as used herein is defined as 'and/or' unless specified otherwise.

The term 'a' or 'an' as used herein is defined as "at least one" unless specified otherwise.

When referring to a noun (*e.g.* a compound, an additive, *etc.*) in the singular, the plural is meant to be included.

In the context of this application, the term 'about' means generally a deviation of 15 % or less from the given value, in particular a deviation of 10% or less, more in particular a deviation of 5%, 4%, 3%, 2%, 1%, 0.5% or less.

The term "substantially" is generally used herein to indicate that it has the general character or function of that which is specified. When referring to a quantifiable feature, this term is generally used to indicate that it is more than 50%, in particular more than 70%, more in particular at least 90%, more in particular at least 95%, even more in particular at least 98% of the maximum of that feature.

Capacitive biosensor

The invention relates to a capacitive biosensor for detecting *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof, comprising a first substrate, a first nucleic acid molecule immobilized on said first substrate and a transcription factor specifically bound to said first nucleic acid molecule, wherein said transcription factor comprises a first binding site comprising the amino acid sequence of SEQ ID NO: 1 or a functional derivative thereof.

A capacitive biosensor is a biosensor that has an ability of detecting a change in a biological system, by detecting a change in capacitance - the change in electric charge with respect to the change in potential - of the device, typically on the surface of a conductive substrate. Said change in capacitance is typically caused by a change in biological environment, such as a change in the accessibility of water to the substrate resulting in a change in (relative) primitivity.

In a system with two substrates (such as electrodes), the capacitance is dependent on three factors: the permittivity of the solution between substrates, the area of the substrates and the distance between the substrates. This is further clarified in formula 1 below.

$$C = \frac{\varepsilon \cdot A}{d} \quad \text{formula 1}$$

wherein, C= capacitance (F), ε = absolute permittivity (F/m), A=area (m²), d = distance between two substrates (m).

Accordingly, a change in capacitance is the result of a change in permittivity. The permittivity describes how much the solution or material between the two substrates polarizes in response to an electric field. In other words, how well the solution or material can store energy. The relative permittivity (ε_r) is the absolute permittivity (ε) divided by the permittivity in vacuum (ε_0).

A capacitive biosensor according to the invention is configured to allow detection of a change in capacitance. An example of such a capacitive biosensor is described by Mannoor et al., Nature Communications, 2012, 3:763.

Said capacitive biosensor is for detection of GHB, a derivative thereof or a salt thereof. Preferably, said capacitive biosensor is for detection of GHB, succinic semialdehyde (SSA) or *gamma*-butyrolactone.

Suitable salts of GHB or a derivative thereof such as SSA or gamma-butyrolactone include sodium salts, potassium salts, ammonium salts, zinc salts,

magnesium salts, calcium salts and the like. Preferably, said suitable salt is a sodium salt.

Substrate

5 A capacitive biosensor according to the invention comprises a first substrate comprising a first nucleic acid molecule immobilized thereon. In principle, any conductive material is suitable to be used as substrate in a capacitive biosensor according to the invention.

10 Preferably, a capacitive biosensor according to the invention comprises a second substrate. Said substrate preferably comprises the same material as said first substrate.

Said first and optionally said second substrate preferably comprises a conductive metal or metal alloy, more preferably selected from gold, silver, copper, iron, aluminum, zinc, nickel, bronze, platinum, lead, brass and (stainless) steel. Most
15 preferably said substrate comprises gold.

Alternatively or additionally, said substrate may comprise a non-metallic material, such as graphene, carbon nanotubes, silicon or a conductive polymer such as polyaniline, polypyrrole and polythiophene.

In a preferred embodiment, a capacitive biosensor according to the invention
20 comprises an interdigitated electrode (IDE). An IDE comprises two electrodes that are interdigitated. This creates a relatively large surface area that is beneficial to measure changes in capacitance. It is beneficial to use a relatively large surface area, because a large surface area reduces the error in the measurements, due to the correlation between capacitance, absolute permittivity and area, as described herein
25 above.

Preferably, said interdigitated electrode comprises a conductive metal or metal alloy, more preferably selected from gold, silver, copper, iron, aluminum, zinc, nickel, bronze, platinum, lead, brass and (stainless) steel. Most preferably said interdigitated electrode comprises a gold interdigitated electrode.

30 Interdigitated electrodes are commercially available, for example an Interdigitated Gold Electrode P-IDEAU50 from Metrohm DropSens may be used.

Transcription Factor

Said capacitive biosensor comprises a transcription factor that is specifically bound to a nucleic acid molecule, and which comprises a first binding site comprising the amino acid sequence

5 TVTLTVREGGEVVYIGCRNSAQPLGHTFRIGMRLPAPFTATGKILLSDLGPGELRM
LFSQFPQPLTSRSVAGLSQLEELALTRARGYSIDDGQIREGMLC (SEQ ID NO:1)
or a functional derivative thereof.

Said first binding site advantageously has an ability of specifically binding *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof. In particular, it is
10 believed that the amino acid residues underlined and in bold are important for binding of said *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof. This has been described in Pan et al., Journal of Biological Chemistry, 2011, 286(23), 20431-20440.

Without wishing to be bound by any theory, it is believed that upon binding of
15 said *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof to said first binding site of said transcription factor, said transcription factor dissociates from said nucleic acid molecule. This dissociation causes a change in capacitance, which may be detected by said capacitive biosensor according to the invention.

Preferably, said transcription factor comprises a second binding site comprising
20 an amino acid sequence of AAELTRFLDLPKSSAHGLLAVMTEL (SEQ ID NO: 2) or a functional derivative thereof.

When said transcription factor is in dimeric or tetrameric form, said second binding site advantageously has an ability of specifically binding said nucleic acid molecule immobilized on said substrate in a capacitive biosensor according to the
25 invention. This has been described in Pan et al., Journal of Biological Chemistry, 2011, 286(23), 20431-20440.

Without wishing to be bound by any theory, it is believed that a transcription factor comprising a first binding site comprising the amino acid sequence of SEQ ID NO: 1 or a functional derivative thereof and a second binding site comprising an
30 amino acid sequence of SEQ ID NO: 2 or a functional derivative thereof undergoes a conformational change upon binding of GHB, a derivative thereof or a salt thereof. This conformational change results in disassociation of the transcription factor from the nucleic acid molecule immobilized on the substrate of a capacitive biosensor according to the invention. This dissociation allows diffusion of water molecules

towards the surface of said substrate, thereby changing the relative permittivity and hence capacitance. The change in capacitance is an indication of the presence of GHB, a derivative thereof or a salt thereof. This is visualized schematically in Figure 8.

With the term “functional derivative” as used herein in relation to a transcription factor is meant a transcription factor that is capable of binding at least substantially to *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof and at least substantially to said nucleic acid molecule immobilized on said substrate. Examples of functional derivatives include mutants of a transcription factor as described herein that have a binding affinity to *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof of at least 1 mM as determinable with isothermal titration calorimetry (ITC). Preferably, in a functional derivative the amino acid residues that are bold and underlined in SEQ ID NO: 1 are conserved.

Preferably, a binding affinity to *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof of at least 100 μ M, at least 10 μ M, at least 1 μ M, most preferably at least 0.8 μ M, as determinable with ITC.

Preferably a functional derivative has a binding affinity to a nucleic acid molecule as defined herein of at least 100 μ M as determinable with isothermal titration calorimetry (ITC). Preferably, a binding affinity to a nucleic acid molecule as defined herein of at least 10 μ M, at least 1 μ M, at least 800 nM, most preferably at least 500 nM, as determinable with ITC.

Excellent results have been obtained with a capacitive biosensor according to the invention, wherein the transcription factor is BlcR, preferably BlcR from *Agrobacterium tumefaciens*, or a functional derivative comprising at least 90% sequence identity thereto, preferably at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, most preferably 100% sequence identity with BlcR, preferably BlcR from *A. tumefaciens*.

As used herein, “percentage of sequence identity” means the value determined by comparing two optimally aligned sequences over a comparison window. For optimal alignment of the two sequences the two sequences are preferably compared over the full length of the shortest of the two sequences. If a polypeptide is compared to a reference sequence, i.e. a bait sequence, the portion of the polypeptide sequence in the comparison window may comprise an addition or a deletion (i.e., a gap) as compared to the reference sequence (which does not comprise an addition or a deletion). The percentage is calculated by determining the number of positions at which an identical

amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison, and multiplying the result by 100 to yield the percentage of sequence identity.

5 BlcR from *A. tumefaciens* (also referred to in the art as *Rhizobium radiobacter* or *Agrobacterium radiobacter*) is a transcription factor (UniProt entry identifier: Q8VPD8) with a total length of 275 amino acid residues, comprising a HTH iclR-type domain located at position 26-87 and a IclR-ED domain located at position 88-270.

The amino acid sequence of BlcR from *A. tumefaciens* is

10 MGQRGQVCQGKCAEDQQSSQISDTVPALRRRAVRILDLVAGSPRDLTAAELTRFL
DLPKSSAHGLLAVMTELDLLARSADGTLRIGPHSLRWANGFLSHLDIVSTFNDHLA
QRHDLDPYTVTLTVREGGEVVYIGCRNSAQPLGHTFRIGMRLPAPFTATGKILLSD
LGPGE LRMLFSQFPQPLTSRSVAGLSQLEELALTRARGYSIDDGQIREGMLCIGA
AIRDYSGAASAGIAISLIRSEASDEKIAYLGEELRTTANALSEKLG YRSQKD (SEQ ID
15 NO: 6).

In solution, BlcR is typically present as a dimer comprising two monomers comprising an amino acid sequence of SEQ ID NO:6 or a functional derivative thereof. Hence, in dimeric form, BlcR comprises two first binding sites for binding of GHB, a derivative thereof or a salt thereof. Further, in dimeric form, BlcR folds such that a
20 binding site is formed capable of binding a first or a second binding site of a nucleic acid molecule as defined herein.

Upon binding of a first nucleic acid molecule, BlcR typically forms a tetramer. In tetrameric form, BlcR thus comprises two second binding sites for binding of a first nucleic acid molecule as defined herein, and four first binding sites for binding of
25 GHB, a derivative thereof, or a salt thereof. This is shown in Figure 7. In the upper panel, binding of a BlcR tetramer to a nucleic acid molecule is shown. In the lower panel, binding of a BlcR dimer to two units of GHB is shown.

Accordingly, in an aspect, a capacitive binding sensor according to the invention comprises a transcription factor specifically bound to said first nucleic acid molecule,
30 wherein said transcription factor is a tetramer, comprising four monomers, each monomer comprising an amino acid sequence of SEQ ID NO:6 or a functional derivative thereof. Such a transcription factor comprises two binding sites for specifically binding said first nucleic acid molecule immobilized on said first

substrate, and four binding sites that are capable of binding GHB, a derivative thereof or a salt thereof.

Accordingly, in a particularly preferred embodiment, the transcription factor comprises an amino acid sequence of SEQ ID NO:6, or a functional derivative thereof.

5 Preferably, said transcription factor is a dimer, comprising two monomers each comprising an amino acid sequence of SEQ ID NO: 6 or a functional derivative thereof, more preferably said transcription factor is a tetramer, comprising four monomers each comprising an amino acid sequence of SEQ ID NO: 6 or a functional derivative thereof.

Nucleic acid molecule

A capacitive biosensor according to the invention comprises a first nucleic acid molecule immobilized on a first substrate.

As used herein, the term “immobilized” refers to covalent binding of said nucleic acid molecule on said substrate.

Said first nucleic acid molecule immobilized on said first substrate may in principle be any nucleic acid molecule that is capable of specifically binding a transcription factor as defined herein above, preferably a transcription factor comprising an amino acid sequence of SEQ ID NO: 2 or a functional derivative thereof, more preferably comprising an amino acid sequence of SEQ ID NO: 6 or a functional derivative thereof.

Preferably, the nucleic acid molecule is a deoxyribonucleic acid (DNA) molecule, in particular double stranded DNA.

Preferably, said nucleic acid molecule comprises between about 20 and about 150 nucleotides per strand, more preferably between about 30 and about 130 nucleotides per strand, even more preferably between about 40 and about 120 nucleotides per strand, in particular between about 50 and about 100 nucleotides per strand. Hence, if said nucleic acid molecule comprises double stranded DNA said nucleic acid molecule preferably comprises between about 20 and about 150 base pairs, more preferably between about 30 and about 130 base pairs, even more preferably between about 40 and about 120 base pairs, in particular between about 50 and about 100 base pairs.

Preferably, said first nucleic acid molecule comprises a first binding site comprising the sequence of ACTCTAATgATTCAAGT (SEQ ID NO: 3) or a functional derivative thereof.

With the term 'functional derivative' in relation to a nucleic acid molecule, is meant a nucleic acid molecule that is capable of binding at least substantially to a transcription factor as defined herein.

Examples of such functional derivatives include mutants of a nucleic acid molecule as described herein that have a binding affinity to a transcription factor as defined herein of at least 1 mM as determinable with isothermal titration calorimetry (ITC).

Preferably a functional derivative of said first nucleic acid molecule has a binding affinity to a transcription factor as defined herein of at least 100 μ M as determinable with isothermal titration calorimetry (ITC). Preferably, a binding affinity to said transcription factor as defined herein of at least 10 μ M, at least 1 μ M, at least 800 nM, most preferably at least 500 nM, as determinable with ITC.

Preferably, in a capacitive binding sensor of the invention, said second binding site of said transcription factor is specifically bound to said first binding site of said first nucleic acid molecule.

Further, said first nucleic acid molecule preferably comprises a second binding site comprising the sequence of a sequence of ATTAGttgaactCTAAT (SEQ ID NO: 4) or a functional derivative thereof.

Preferably, in a capacitive binding sensor of the invention, said second binding site of said transcription factor is specifically bound to said second binding site of said nucleic acid molecule.

Particularly preferred herein is a first nucleic acid molecule comprising a nucleic acid sequence comprising said first and said second binding site linked by a spacer. Said spacer is typically about 3 nucleotides long. More preferably the spacer has a sequence of tca.

Excellent results have been achieved with a first nucleic acid molecule comprising a nucleic acid sequence of ccatagttcACTCTAATGATTCAAGTtcaATTAGttgaactCTAATgcggg (SEQ ID NO: 5). Herein, said first nucleic acid sequence comprises said first binding site comprising the nucleic acid sequence of SEQ ID NO: 3 and said second binding comprising the nucleic acid sequence of SEQ ID NO: 4 linked by a spacer having the sequence tca.

Advantageously, in such a capacitive binding sensor according to the invention, said first nucleic acid molecule comprising a nucleic acid sequence of SEQ ID NO: 5 is capable of specifically binding two dimers of said transcription factor. Typically, said nucleic acid molecule comprises a first binding site specifically bound to a second
 5 binding site of a first dimer of said transcription factor and a second binding site specifically bound to a second binding site of said second dimer of the transcription factor.

As the skilled person will appreciate, if said first nucleic acid molecule is double stranded DNA, said first nucleic acid molecule will comprise a first strand comprising
 10 a nucleic acid sequence of SEQ ID NO:3, SEQ ID NO:4 or SEQ ID NO:5 or a functional derivative thereof, and a second strand comprising a nucleic acid sequence complementary to said first strand. The term “complementary DNA strand” is well known in the art to refer to a strand of DNA that is capable of specifically hybridizing to said first strand. Typically, a complementary DNA strand comprises nucleotides
 15 arranged in an order in the sequence such that most nucleotides are capable of forming a base pair with said nucleotides of said first strand.

Preferably, said complementary DNA strand comprises a sequence of nucleotides in an order such that each nucleotide is capable of forming a base pair with the nucleotides of said first strand. However, the skilled person will appreciate
 20 that some mismatched base pairs may be present in said double stranded DNA, whilst still maintaining its double-stranded structure. Typically, about at most 20% of mismatched base pairs is tolerable, preferably about at most 15%, at most 10%, at most 8%, at most 6%, at most 4%, at most 2% of mismatched base pairs are present in said double stranded DNA. Said percentage may be calculated by dividing the number
 25 of mismatches by the total number of base pairs, and multiplying by 100.

Preferably, said first nucleic acid molecule comprises a first strand comprising a nucleic acid sequence of SEQ ID NO: 3 and a second strand comprising a nucleic acid sequence of ACTTGAATCATTAGAGT (SEQ ID NO: 7).

Alternatively or additionally, said first nucleic acid molecule comprises a first
 30 strand comprising a nucleic acid sequence of SEQ ID NO: 4 and a second strand comprising a nucleic acid sequence ATTAGAGTTCAACTAAT (SEQ ID NO: 8).

Particularly preferred is a first nucleic acid molecule comprising a first strand comprising a nucleic acid sequence of SEQ ID NO: 5 and a second strand comprising a

nucleic acid sequence

CCCGCATTAGAGTTCAACTAATTGAACTTGAATCATTAGAGTGAACATATGG

(SEQ ID NO: 9).

5 Preferably, a capacitive biosensor according to the invention comprises a second nucleic acid molecule immobilized on a second substrate. Said second nucleic acid molecule is different from said first nucleic acid molecule, preferably a nucleic molecule that is not capable of specifically binding a transcription factor as defined herein.

10 Preferably, said second nucleic acid molecule is a DNA molecule, more preferably double-stranded DNA molecule. Preferably, said sequence comprises the nucleic acid sequence of
ATACTGCAATGTACTTAATAGATCACCGCTGCTTGCACTAGTGGTTATAAT (SEQ ID NO: 10).

15 More preferably, said second nucleic acid molecule comprises a first sequence of SEQ ID NO: 10, specifically hybridized to a second nucleic acid sequence that is complementary to said first nucleic acid sequence. Preferably, said second nucleic acid molecule comprises the nucleic acid sequence of
ATTATAACCACTAGTGCAAGCAGCGGTGATCTATTAAGTACATTGCAGTAT (SEQ ID NO: 11).

20 Advantageously, a capacitive biosensor according to the invention comprising a second nucleic acid molecule immobilized on a second substrate allows more accurate detection of GHB, a derivative thereof or a salt thereof. By comparing capacitance between the first and second substrates, any changes in capacitance caused by factors other than dissociation of the transcription factor from the first nucleic acid molecule
25 immobilized on the first substrate (e.g. salt concentrations), may be subtracted. Hence, said second substrate comprising said second nucleic acid molecule immobilized thereon may thus function as a reference.

Said first nucleic acid molecule and optionally said second nucleic acid molecule may be immobilized on said first and said second substrate using any suitable manner
30 known in the art that allows covalent binding of nucleic acids to a substrate, preferably gold.

Preferably, said nucleic acid molecule is thiol modified and directly modified to said substrate, preferably a gold substrate, more preferably a gold interdigitated electrode.

Preferably, said substrate further comprises mercaptohexanol to mask unreacted parts of the substrate surface.

The concentration of nucleic acid molecules immobilized on said substrates is typically selected such that a change in capacitance is optimal. A too high

5 concentration of nucleic acid molecules will result in a situation in which the surface is too crowded for BlcR to bind to the first nucleic acid molecule. With an optimal concentration, the BlcR will be able to bind to the first nucleic acid molecule and the change in capacitance will be high enough to be measured. Typical concentrations of first nucleic acid molecules when contacted with said first substrates are between 1
10 μM and between 1 mM, preferably between about 10 μM and about 100 μM . It has been found that with such concentrations, a final concentration of first nucleic acid molecules immobilized on said first substrates is obtained that allows proper measuring of capacitance.

15 In a particularly preferred embodiment, the invention relates to a capacitive biosensor, comprising an interdigitated gold electrode comprising a first and a second electrode, further comprising

- a first nucleic acid molecule immobilized on said first electrode and a transcription factor specifically bound to said first nucleic acid molecule, wherein said transcription
20 factor comprises a first binding site comprising the amino acid sequence of SEQ ID NO: 1 or a functional derivative thereof, more preferably wherein the transcription factor comprises the amino acid sequence of SEQ ID NO: 6 or a functional derivative thereof; and

- a second nucleic acid molecule immobilized on said second electrode, said second
25 nucleic acid molecule being different from said first nucleic acid molecule, preferably wherein said second nucleic acid molecule does not comprise a transcription factor specifically bound to said second nucleic acid molecule.

Preferably herein, said first nucleic acid molecule comprises the nucleic acid sequence of SEQ ID NO: 5. Preferably said second nucleic acid molecule comprises the
30 nucleic acid sequence of SEQ ID NO: 10.

Device

The invention further relates to a device for detecting *gamma*-hydroxybutyric acid or a derivative thereof, comprising a capacitive biosensor according to any one of the preceding claims configured to detect *gamma*-hydroxybutyric acid in a liquid.

5 Preferably, said device is a container, preferably a container for holding a beverage, such as a cup, a glass, a bottle, a mug or a jug.

Said device comprises a capacitive biosensor according to the invention further comprising an electrical device configured to become activated when critical levels of *gamma*-hydroxybutyric acid are detected.

10 In the context of the present application, with the term 'critical levels of GHB, a derivative thereof or a salt thereof' is meant a concentration that exceeds levels of GHB that may be naturally present in said liquid, such as in a beverage. It is known in the art that certain beverages may contain small amounts of GHB, which is not harmful to the consumer. (Elliott et al. Forensic Science International, 2005. 151(2-
15 3):289-292).

Accordingly, preferably, said liquid comprises at least 25 mg/L GHB, a derivative thereof, or a salt thereof, more preferably at least 40 mg/L, at least 50 mg/L, most preferably at least 75 mg/L of GHB, a derivative thereof, or a salt thereof.

Preferably, the concentration of GHB, a derivative thereof or a salt thereof is
20 between about 25 mg/L and about 1000 mg/L, more preferably between about 40 mg/L and about 800 mg/L, even more preferably between about 50 mg/L and about 600 mg/L.

Said electrical device may be any electrical device that is capable of providing a measurable signal. Preferably, said electrical device is a light emitter, a sound emitter
25 or a Bluetooth transmitter.

Such a device advantageously allows passive detection of GHB, a derivative thereof or a salt thereof. In presence of GHB, a derivative thereof or a salt thereof, said electrical device becomes activated and immediately warns the owner of the device of the presence of critical levels of GHB. The device advantageously is easy to
30 use, and passive, meaning that no further steps are required to detect GHB, i.e. it is not required to take a sample from a liquid for testing and contact it with the device, as all relevant components for detection of GHB are integrated in the device.

Furthermore, the device advantageously allows multiple use. This means that once GHB, a derivative thereof or a salt thereof is detected, the system can be brought

back to equilibrium, for example by removing the liquid and washing the device, and can be used again for another cycle of GHB detection. By removing the liquid containing the GHB, the transcription factor again becomes specifically bound to the nucleic acid molecule and the device is ready to be used again.

5 Said liquid may be any liquid that may comprise critical levels of GHB, a derivative thereof or a salt thereof. Preferably said liquid is a body fluid or a beverage, more preferably a beverage. Examples of beverages suitable for detection of GHB include soda's, juices, water, dairy and alcoholic beverages including spirits such as tequila and rum, and other alcoholic beverages such as wine, beer and the like.

10 Examples of bodily fluids suitable for detection of GHB include urine, saliva, blood, plasma, serum or sweat. Preferably, the body fluid is urine.

Preferably, said liquid is a beverage. This allows the device to directly detect the presence of GHB and prevent its consumption and negative consequences associated with GHB consumption.

15 Alternatively, if the liquid is a body fluid, said device can detect the presence of GHB, a derivative thereof or a salt thereof after consumption of said GHB, derivative thereof or salt thereof, and provide proof of consumption of said substance. This is in particular relevant in case GHB has been consumed against a subject's will and/or without the subject's consent.

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Method for detecting *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof

The invention further relates to a method for detecting *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof, comprising the steps of:

- 25 - obtaining a first value of capacitance of a capacitive biosensor according to the invention or a device according to the invention;
- contacting said capacitive biosensor or said device with a liquid comprising *gamma*-hydroxybutyric acid, a derivative thereof, or a salt thereof;
- obtaining a second value of capacitance of said capacitive biosensor or said device
- 30 after said contacting; and
- detecting a change in capacitance by subtracting said second value from said first value.

Said capacitive biosensor, said device and said liquid are as defined herein above.

Preferably, said change is a positive value indicating that said first value is lower than said second value. This typically indicates diffusion of water molecule towards the substrate, due to dissociation of said transcription factor from said nucleic acid molecule.

5 Preferably, said change in capacitance results in activation of an electrical device, preferably a light emitter.

For good results, typically a threshold value of GHB, a derivative thereof or a salt thereof is used, to avoid detection of GHB that may be naturally present in a liquid, such as in a beverage.

10 Accordingly, the invention preferably relates to a method wherein the presence of *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof is detected when the difference is more than 0.1 pF, preferably more than 0.05 pF, more preferably more than 0.02 pF. Said difference may be determined using any suitable method known in the art, for example using a universal transducer interface or using a vector
15 impedance, as described in the Examples below.

Preferably, said change in capacitance corresponds to a concentration of GHB, a derivative thereof or a salt thereof present in a liquid of at least 50 mg/L, more preferably at least 60 mg/L, more preferably at least 75 mg/L, even more preferably at least 100 mg/L.

20 Preferably, said change in capacitance corresponds to a concentration of GHB, a derivative thereof, or a salt thereof present in a liquid wherein said *gamma*-hydroxybutyric acid or derivative thereof is present in said liquid in a concentration of between about 50 mg/L and about 1000 mg/L, preferably between about 75 mg/L and about 500 mg/L, most preferably between about 100 mg/L and about 250 mg/L.

25 These are typical concentrations that may be found in a liquid, preferably a beverage, when GHB, a derivative thereof or a salt thereof is administered into the liquid.

A method according to the invention is advantageously fast in detecting GHB, a derivative thereof or a salt thereof. This is important to prevent consumption of a
30 liquid, such as a beverage wherein GHB, a derivative thereof or a salt thereof is present.

Preferably, in a method according to the invention said change in capacitance is detected between about 1 second and about 10 minutes after said contacting, preferably between about 10 seconds and about 5 minutes after said contacting, more

preferably between about 30 seconds and about 2 minutes after said contacting, most preferably said change in capacitance is detected essentially immediately after said contacting.

5 The invention further relates to a use of a capacitive biosensor or a device according to the invention for detection of *gamma*-hydroxybutyric acid in a liquid, preferably a beverage.

For the purpose of clarity and a concise description, features are described herein as part of the same or separate embodiments, however, it will be appreciated that the scope of the invention may include embodiments having combinations of all
10 or some of the features described.

The invention is demonstrated by the following examples.

Examples

Example 1: Preparation of a capacitive biosensor according to the invention

15 The following materials were obtained:

- P-IDEAU50 from Metrohm DropSens
- Thiol modified single strand oligonucleotides from Eurofins Genomics with sequences
 - 20 - *[ThiolC6]CCATAGTTCACTCTAATGATTCAAGTTCAATTAGTTGAAC TCTAATGCGGG* (HPLC purified), referred to as oligonucleotide A1 in this protocol
 - *CCCGCATTAGAGTTCAACTAATTGAACTTGAATCATTAGAGTGAAC TATGG* (Salt free), referred to as oligonucleotide A2 in this protocol
 - *[ThiolC6]ATACTGCAATGTACTTAATAGATCACCGCTGCTTGCACTA*
25 *GTGGTTATAAT* (HPLC purified), referred to as oligonucleotide B1 in this protocol
 - *ATTATAACCACTAGTGCAAGCAGCGGTGATCTATTAAGTACATTGC AGTAT* (Salt free), referred to as oligonucleotide B2 in this protocol
- Milli-Q water
- 30 - $\geq 99.5\%$ 3-(*N*-morpholino)propanesulfonic acid (MOPS) from Calbiochem
- $\geq 99.5\%$ MgSO_4 from Sigma Aldrich
- 97% mercaptohexanol (MCH) from Sigma Aldrich
- Na_2SO_4 from Sigma Aldrich
- 99.999% H_2SO_4 from Sigma Aldrich

- NaOH from Calbiochem
- Isopropanol from Sigma Aldrich
- 97% dithiothreitol (DTT) from Sigma Aldrich
- $\geq 99.5\%$ Ethyl acetate from Sigma Aldrich
- 5 - Succinyl-semialdehyde (SSA) from Santa Cruz Biotechnology

In order to measure the change in capacitance caused by SSA addition and thus BlcR binding and unbinding, a two-electrode system has been introduced. On electrode A, oligonucleotides have been immobilized that contain the wildtype BlcR binding site. Electrode B contains scrambled oligonucleotides, thus without the BlcR binding site.

First, the buffers are prepared. The MOPS buffer contained 20 mM MOPS and 30 mM sodium sulfate (Na_2SO_4) in Milli-Q water. The MOPS Mg buffer contained 2.5 mM Magnesium sulfate (MgSO_4), 20 mM MOPS and 30 mM sodium sulfate (Na_2SO_4) in Milli-Q water. Both buffers have been set to pH=7 with H_2SO_4 and NaOH.

Secondly, the IDEs have been cleaned with sonication. The electrodes were covered with isopropanol and put in a sonication bath for 5 minutes. After this, the electrodes have been thoroughly washed with Milli-Q water.

Then the first, thiol modified oligonucleotide strands (A1 and B1) have been reduced and immobilized to respectively electrode A and B based on the protocol for thiol modified oligonucleotides reduction by Integrated DNA Technologies. The thiol on the strands can be oxidized and can form disulfide interactions. However, to let the thiol bind to the IDE, the disulfide bridges need to be broken and therefore the strands need to be reduced. A stock solution of 100 mM DTT in MOPS buffer and a stock solution of 100 μM of every oligonucleotide has been made. Then 1 μL of DTT stock solution has been taken and combined with 10 μL of the stock solution of A1 in one PCR tube. 1 μL of DTT stock solution has been taken and combined with 10 μL of the stock solution of B1 in another PCR tube. Both tubes have been vortexed shortly and been incubated for 15 minutes. Then 11 μL of ethyl acetate was added to both tubes and vortexed again. Both tubes were centrifuged shortly. After this, the solution is separated into two layers. The upper layers (with DTT and ethyl acetate) have been removed from the bottom layers (oligonucleotide in MOPS). The addition of ethyl acetate and removal with centrifugation after vortexing has been repeated three more times to both tubes. After the upper layers have been removed for the last time, the

bottom layer of both tubes have directly been combined with 990 μ L MOPS buffer and directly been incubated on the electrodes to prevent thiol oxidation again. The electrodes have been incubated overnight at room temperature.

After the incubation of strand A1 and B1, the electrodes are washed thoroughly to remove all unbound strands. This was done by holding the electrodes with a tweezer under an angle and placing a piece of tissue under the electrode. 2 mL of MOPS buffer then has been pipetted very carefully from the top of the electrode and letting the buffer run over the whole electrodes.

Then MCH was added to the electrodes. MCH inactivates the rest of the gold that has not been covered by the thiol modified oligonucleotides, so no other molecules can bind to the electrode. A stock solution of 1 mM MCH in MOPS buffer has been made, of which 1 mL was incubated on an electrode. This solution has been incubated for 1 hour. After incubation with MCH, the electrodes have been washed again in the same way as described above.

Now that the rest of the surface was deactivated with MCH, the complementary strands, A2 and B2, have been added to electrode A and B respectively. 10 μ L of the 100 μ M stock solutions of A2 has been combined with 990 μ L MOPS buffer and added to the electrode. The same has been done for oligonucleotide B2. These solutions have been incubated overnight at room temperature.

After both oligonucleotide strands have been immobilized, the electrodes have been washed again with the steps described above.

Example 2: Vector impedance measurements

After DNA immobilization, the capacitive biosensor of Example 1 has been analyzed with vector impedance measurements. A small cut has been made in the plastic between the arms of the electrodes in order for the electrode to fit in the meter without the need of extra cables. In this case, an Agilent 4294A Precision Impedance Analyzer (40 Hz - 110 MHz) has been used. The marker was set at 1 kHz, since at this frequency the capacitance was relatively stable but the resistance was still relatively high.

The capacitance has been measured while no BlcR was added to the electrodes. The electrodes have been put in a bath of MOPS Mg buffer. The capacitance of electrode A and B at 1 kHz was measured (experiment 1A for electrode A, 1B for electrode B). Then, the electrodes have been incubated in a bath with 1 mL of 9.5 μ M

BlcR in Mg MOPS buffer for 1 hour. The capacitance at 1 kHz has been measured in this bath (experiment 2A for electrode A, 2B for electrode B). Lastly, 10 μ L of a 15 mM SSA stock solution in Mg MOPS has been added to the BlcR bath. The capacitance at 1 kHz has been measured again (experiment 3A for electrode A, 3B for electrode B).

- 5 The capacitance in all experiments was measured five times, after which the average was taken.

The results are presented in table 1 and figure 1.

- 10 Table 1. Results of the average capacitance during the vector impedance measurements. The capacitance of the electrode A (with BlcR binding site on DNA) and electrode B (without BlcR binding site on DNA) has been measured without BlcR or SSA addition, labeled '1A' for electrode A and '1B' for electrode B, after incubation with BlcR, labeled '2A' for electrode A and '2B' for electrode B, and after incubation with SSA, labeled '3A' for electrode A and '3B' for electrode B. The standard deviation
- 15 has also been included.

Experi ment	First capacitanc e measur ment (pF)	Second capacitanc e measur ment (pF)	Third capacitanc e measur ment (pF)	Fourth capacitanc e measur ment (pF)	Fifth capacitanc e measur ment (pF)	Average capacita nce (pF)	Standa rd deviati on
1A	1,156	1,177	1,242	1,150	1,249	1,195	0,048
2A	0,734	0,871	0,722	0,865	0,824	0,803	0,071
3A	0,868	0,906	0,848	0,947	0,926	0,899	0,041
1B	1,076	1,136	1,144	1,028	1,020	1,081	0,058
2B	1,102	0,978	1,009	1,126	1,067	1,056	0,062
3B	0,950	1,001	0,960	1,101	0,995	1,002	0,060

In figure 1, it can be seen that with electrode A, the capacitance goes down after addition of BlcR, which indicates that the BlcR is bound to DNA, after addition of SSA

the capacitance goes up again which indicates binding of SSA to BlcR followed by dissociation of BlcR from the DNA. It can also be derived from figure 1 and table 1 that the SSA addition does not remove all BlcR, since the capacitance does not reach the initial capacitance value. With electrode B, we can see that capacitance does not change significantly after BlcR or SSA addition.

These results show that the presence of SSA can be detected by a capacitive biosensor according to the invention

Example 3: Universal transducer interface measurements

Next to the vector impedance measurements, universal transducer interface (UTI). UTI was used to measure the capacitance more precisely. A UTI measures the unknown capacitance by using two reference capacitances that are precisely known. A picture of the UTI board is shown in figure 2 and a diagram of the electrical system is shown in figure 3.

The system first converts the capacitance to a voltage using a capacitance to voltage converter (CVC). This CVC builds up charge over the capacitor and unloads it, building up a square wave voltage signal. This voltage signal is converted into a specific period using a voltage-to-period convertor (VPC). These conversions are done for the unknown and the two reference capacitors, guided by the multiplexers. A digital divider is used to change the frequency to the desired data acquisition frequency. An external control system is used.

Before using the board, the pins were soldered to the board to be able to connect the wires. A custom UTI software created by employees of the TU Delft. The UTI was connected to a laptop with a USB to USB-A cable. Mode 1 (C23: 3 capacitors 0-2pF) was used for the measurements. The pin connections were as follows:

- Pin C was connected to a reference capacitor of 1,8 pF.
- Pin D was connected to our electrode by making use of part of a cable (Ref. CACIDEP).²
- Pin A was connected to the outputs of the capacitors from C and D.
- Pin B, E, and F were not connected.

This is visualized in figure 4.

Inside the software, the reference capacitance was changed to 1,8pF and the average index to 50. The rest of the settings remained the initial conditions. An overview of the settings and an image of the software is shown in figure 5.

The electrode was placed in a tube containing 1mL MOPS Mg buffer while making sure that the fingers were fully covered. The average capacitance of 50 measurements was noted. Then 0,25 ml of buffer was then removed and 0,25 ml of 18.55 μ M BlcR in MOPS Mg was added. After an hour of incubation, an average of 50 capacitance measurements was noted. Lastly, 10 μ l of a 15 mM SSA stock solution was added. After 10 minutes of incubation, an average of 50 capacitance measurements was noted. The results are presented in table 2 and figure 6.

Table 2. Results of the average capacitance during the UTI measurements. The capacitance of the electrode A (with BlcR binding site on DNA) and electrode B (without BlcR binding site on DNA) has been measured without BlcR or SSA addition, labeled '1A' for electrode A and '1B' for electrode B, after incubation with BlcR, labeled '2A' for electrode A and '2B' for electrode B, and after incubation with SSA, labeled '3A' for electrode A and '3B' for electrode B.

Experiment	Average capacitance (pF)
1A	0,7129
2A	0,5602
3A	0,6421
1B	0,8672
2B	0,8677
3B	0,8682

It can be derived from a comparison of Figures 1 and 6 that the results of the vector impedance measurements (Example 2) and UTI measurement (Example 3) are comparable. The capacitance of electrode A lowers when BlcR is introduced and increases again when SSA is added. For electrode B, the capacitance does not vary

significantly when BlcR or SSA is added. This supports that BlcR binding decreases the capacitance and addition of SSA increases the capacitance when DNA with a BlcR binding site is immobilized. This supports that a system can be created in which the capacitance is changed when SSA is added, while the effects of other conditions (such as temperature or salt concentration changes) are minimized by comparing electrode A and B. The changes caused by these other conditions will change the capacitance of both electrodes, while SSA addition changes only the capacitance of electrode A. By subtracting the capacitance of electrode B of the capacitance of electrode A, only the capacitance change caused by SSA addition can be measured.

Citations

1. https://sfvideo.blob.core.windows.net/sitefinity/docs/default-source/protocol/reduction-protocol-for-thiol-modified-oligonucleotides.pdf?sfvrsn=431c3407_8; accessed on 27 September 2022
2. https://dropsens.com/pdfs_productos/new_brochures/cac-bicac-cac8x.pdf; accessed on 27 September 2022.

Claims

1. A capacitive biosensor for detecting *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof, comprising a first substrate, a first nucleic acid molecule
5 immobilized on said first substrate and a transcription factor specifically bound to said first nucleic acid molecule, wherein said transcription factor comprises a first binding site comprising the amino acid sequence of SEQ ID NO: 1 or a functional derivative thereof.
2. The capacitive biosensor according to claim 1, wherein the transcription factor
10 further comprises a second binding site comprising an amino acid sequence of SEQ ID NO: 2 or a functional derivative thereof.
3. The capacitive biosensor according to claim 1 or 2, wherein the transcription factor is BlcR, preferably BlcR from *Agrobacterium tumefaciens*, or a functional derivative thereof, preferably comprising at least 95% sequence identity thereto.
- 15 4. The capacitive biosensor according to any one of the preceding claims, wherein said nucleic acid molecule comprises a first binding site comprising the sequence of SEQ ID NO: 3 or a functional derivative thereof.
5. The capacitive biosensor according to any one of the preceding claims, wherein said nucleic acid molecule comprises a second binding site comprising the sequence of
20 SEQ ID NO: 4 or a functional derivative thereof.
6. The capacitive biosensor according to any one of the preceding claims, wherein said nucleic acid molecule comprises a sequence of SEQ ID NO: 5 or a functional derivative thereof.
7. The capacitive biosensor according to any one of the preceding claims, further
25 comprising a second nucleic acid molecule immobilized on a second substrate, said second nucleic acid molecule being different from said first nucleic acid molecule, preferably wherein said second nucleic acid molecule does not comprise a transcription factor specifically bound to said second nucleic acid molecule.
8. The capacitive biosensor according to any one of the preceding claims,
30 comprising an interdigitated electrode, preferably an interdigitated gold electrode.
9. The capacitive biosensor according to any one of the preceding claims further comprising an electrical device configured to become activated when *gamma*-hydroxybutyric acid is detected.

10. The capacitive biosensor according to claim 9, wherein the electrical device is a light emitter, a sound emitter or a Bluetooth transmitter.

11. A device for detecting *gamma*-hydroxybutyric acid or a derivative thereof, comprising a capacitive biosensor according to any one of the preceding claims
5 configured to detect *gamma*-hydroxybutyric acid in a liquid.

12. The device according to claim 11, wherein the device is a container, preferably a container for holding a beverage, such as a cup, a glass, a bottle, a mug or a jug.

13. A method for detecting *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof, comprising the steps of:

- 10 - obtaining a first value of capacitance of a capacitive biosensor according to any one of claims 1 to 10 or a device according to any one of claims 11 or 12;
- contacting said capacitive binding sensor or said device with a liquid comprising *gamma*-hydroxybutyric acid, a derivative thereof, or a salt thereof;
- obtaining a second value of capacitance of said capacitive biosensor or said device
15 after said contacting; and
- detecting a change in capacitance by subtracting said second value from said first value.

14. The method according to claim 13, wherein the presence of *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof is detected when said
20 second value is higher than said first value, preferably when the absolute value of the difference is more than 0.1 pF.

15. The method according to any one of claims 13 or 14, wherein said change in capacitance results in activation of an electrical device, preferably a light emitter.

16. The method according to any one of claims 13 to 15, wherein said liquid is a
25 beverage or a body fluid, preferably urine, blood, serum, plasma, saliva or sweat.

17. The method according to any one of claims 13 to 16, wherein said derivative is succinic semialdehyde or *gamma*-butyrolactone.

18. The method according to any one of claims 13 to 17, wherein said *gamma*-hydroxybutyric acid, derivative thereof or salt thereof, is present in said liquid in a
30 concentration of between about 50 mg/L and about 1000 mg/L, preferably between about 75 mg/L and about 500 mg/L, most preferably between about 100 mg/L and about 250 mg/L.

19. The method according to any one of claims 13 to 18, wherein said change in capacitance is detected between about 1 second and about 10 minutes after said

contacting, preferably between about 10 seconds and about 5 minutes after said contacting, more preferably between about 30 seconds and about 2 minutes after said contacting, most preferably said change in capacitance is detected essentially immediately after said contacting.

- 5 20. Use of a capacitive biosensor according to any one of claims 1-10 or a device according to any one of claims 11 or 12 for detection of *gamma*-hydroxybutyric acid in a liquid, preferably a beverage.

Title: capacitive biosensor for detection of GHB

Abstract

The invention relates to a capacitive biosensor for detecting *gamma*-hydroxybutyric acid, a derivative thereof or a salt thereof, comprising a first substrate, a first nucleic acid molecule immobilized on said first substrate and a transcription factor specifically bound to said first nucleic acid molecule, wherein said transcription factor comprises a first binding site comprising the amino acid sequence of SEQ ID NO: 1 or a functional derivative thereof.